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Correspondent Banking, Systemic Risk, and the Panic of 1893

During the U.S. National Banking Period (1863–1913), a network of correspondent bank relationships created vulnerabilities to bank failures and financial panics. Using data on correspondent relationships for all national, state, savings, and private banks just before the Panic of 1893, along with the precise dates of bank suspensions, we show that prior suspensions of both upstream and downstream correspondents increased the likelihood that a given bank would itself suspend, and that these effects varied over the Panic. Conditional on suspension, banks with prior correspondent suspensions were also more likely to reopen. New York Clearinghouse banks, despite low incidences of actual failure, saw significant balance sheet weakening early in the Panic when downstream respondents suspended, and falling stock prices throughout.

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For much of its history, the U.S. banking system consisted of fragmented unit banks operating in single locations. This structure required banks to form complex correspondent networks to facilitate the movement of capital and the clearing of payments across space and time. Although these relationships enabled the growth of a nationwide financial system during the National Banking period (1863–1913), at times they also served as conduits for transmitting financial shocks, contributing to the system's instability (see, e.g., Bordo, Rockoff, and Redish 1996). The Panic of 1893 offers an opportunity to examine the role of correspondent networks during a major systemic event. As in the Great Depression, the Panic of 1893

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emanated from the interior rather than New York City, which stood at the apex of the system's pyramid of reserves, and the vast majority of bank suspensions, numbering more than 550 in total, occurred outside of New York. The early phase, starting in late April, saw flights to liquidity in interior cities and a draining of reserves from the central reserve cities of New York and Chicago, while a partial suspension of cash payments to the interior by member banks of the New York Clearinghouse (NYCH) on August 3 put a later phase into motion, with downstream banks increasingly unable to recover reserves from upstream correspondents. The Panic thus facilitates study of transmission both up and down the network.

We examine the role of bank suspensions in the Panic's transmission using data on the entire network during the disruption. By suspension, we mean an inability of a given bank to redeem its deposit liabilities in cash, leading it to "suspend" operations but not necessarily to close permanently. We refer to a bank that holds interbank deposits of other banks in the network as a "correspondent," and the banks placing these interbank deposits as its "respondents." In this setting, a given bank can have no interbank connections at all, can be either a correspondent or a respondent, or can serve as a correspondent for some downstream banks and at the same time be a respondent of others upstream. Our data include more than 12,500 banks along with the identities of their primary correspondents collected from the 1893 edition of Rand-McNally and Company's *Bankers' Directory and List of Bank Attorneys*, which lists all national, state, savings, and private banks. By using this set of correspondent connections to construct the full network, our unique data set extends earlier studies of 1893 by tracing the impact of a prior bank suspension on both upstream correspondents and downstream respondents, differentiating between national banks and other types of banks. Data on bank suspensions are from various issues of *Bradstreet's* (1893).

We find that correspondent suspensions substantially increased the probability of subsequent suspensions of their respondents, and similarly that respondent suspensions increased the probability their correspondents would later suspend. We establish this notion of Granger causality for both the upstream and downstream transmission of bank suspensions using the precise date of each suspension. Moreover, the importance of correspondent and respondent suspensions disappears later in the Panic following a partial suspension of payments to the interior by members of the NYCH, suggesting that this disruption of the broader interbank network diminished the importance of failures within a bank's particular set of correspondents. We also find that suspended banks with a previously suspended correspondent were more likely to reopen than banks suspending for other reasons, suggesting that correspondent suspensions represented primarily liquidity rather than solvency events.

We also collect weekly balance sheet and stock price data to study the impact of respondent suspensions on New York City banks. Moving beyond suspensions as a measure of financial stress allows us to understand better what was happening in the central node of the network, given that New York experienced only four bank suspensions. The balance sheets for the NYCH banks come from weekly issues of the *Commercial and Financial Chronicle*, and our empirics demonstrate that respondent suspensions led to significant declines in loans, deposits, and specie reserves of New

York banks in the early stage of the Panic. Since the NYCH ceased publication of balance sheet items after June 10 to forestall runs on vulnerable New York banks, we also use the published stock prices of its members from the *Chronicle* to show that New York banks more exposed to respondent suspensions saw larger declines in their stock prices throughout the disruption.

One example of upstream weakness and an accompanying stock price response during the "blackout" period involves the St. Nicholas Bank in New York, which ultimately failed in December 1893. Its path to demise began with the August failure of the Madison Square Bank—another New York bank—for which St. Nicholas was the clearing agent. The stock price of St. Nicholas on August 8, the day before announcing it would no longer clear transactions for Madison Square, was $125. The stock price fell to $113.50, and deposits fell 19% over the next two months as details of the problems at Madison Square and the role of St. Nicholas as a $267,000 creditor became public. Soon after, the price of St. Nicholas fell dramatically as depositors withdrew funds, leading to its closure on December 21 with heavy losses (*New York Times*, various issues).

Previous work by Calomiris and Carlson (2017) provides an analysis of the 1893 panic that is particularly relevant to ours. They study the role of the interbank network on suspensions among 208 national banks in 38 major cities using reports of bank examiners. These reports provide the amount of deposits a given bank had with each of its upstream reserve agents, which in turn reveal that greater exposure to upstream correspondents through interbank deposits related to higher rates of bank suspension. We complement their contribution by broadening the scope of analysis beyond national banks to the full banking network, albeit not to the same degree of detail, examining both upstream and downstream transmission of financial shocks, and studying the outcomes of New York banks specifically through weekly balance sheet and stock price data.

More broadly, our study contributes to a growing literature on the role of banking networks in transmitting financial shocks. A number of articles offer theoretical frameworks for how the structure of interbank relationships can affect financial fragility, including Allen and Gale (2008), Gai, Haldane, and Kapadia (2011), and Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015). Much of the empirical work on the role of correspondent networks has focused on the Great Depression. Das, Mitchener, and Vossmeyer (2022) demonstrate that the three-tiered pyramid structure of interbank relationships made the banking system less stable than it otherwise could have been in 1929. Mitchener and Richardson (2019), using quarterly data at the Federal Reserve district level for Fed member banks, show that respondent suspensions during the Great Depression led to withdrawals of interbank deposits from reserve and central reserve cities, leading these banks to curtail lending significantly. Like Mitchener and Richardson, we also measure the impact of respondent suspensions on outcomes beyond bank suspensions.

Calomiris, Jaremski, and Wheelock (2022) collect correspondent data from the Rand-McNally directory for all national banks from 1929 to 1934, and find that the suspensions of both correspondents and respondents increased a given national bank's likelihood of suspension. They conclude that what they call "contractual contagion" was a key factor in the bank failures of the Great Depression. We broaden their approach to include all banks in a study of systemwide contagion and bank suspensions, and exploit the timing of bank suspensions to establish more proximate causation from suspension of a given bank to subsequent suspensions of its correspondents or respondents. Our main contribution, therefore, is to provide an analysis of interbank contagion effects that accounts for the precise timing of suspensions with a unique data set covering all reporting national, state, and private banks for a significant banking panic of the Gilded Age, and in particular during a time when the U.S. banking system lacked some of the guardrails put in place by the Federal Reserve Act of 1913.

Section 1 provides background on the correspondent system and the Panic of 1893. Section 2 presents results on the role of prior correspondent and respondent suspensions in determining the likelihood of a given bank's failure during the Panic. Section 3 investigates the impact of respondent suspensions on the balance sheets and stock valuations of New York banks. Section 4 concludes.

1. THE CORRESPONDENT BANKING SYSTEM AND THE PANIC OF 1893

The network of correspondent relationships in the U.S. banking system derived from its fragmented nature and lack of branching. Unit banks located outside of financial centers sought out correspondents in larger cities to serve the needs of their customers more effectively. The National Bank Acts of 1863 and 1864 cemented these relationships by designating certain cities as reserve or central reserve cities, and by allowing deposits with reserve city correspondents to count toward a bank's reserve requirements. Anderson, Paddrik, and Wang (2019), in a comparison of Pennsylvania banks in 1862 and 1867, demonstrate that the Acts concentrated interbank deposits at the city level and in particular banks, leading to systematically important banks and a greater likelihood of financial contagion. We study these types of interdependencies using the full set of primary correspondent relationships for all national, state, and private banks operating in January 1893. For a typical bank in our sample, the *Bankers' Directory* lists a New York City correspondent and another in a geographically proximate reserve or other central reserve city (i.e., Chicago or St. Louis), although a few banks listed as many as six primary correspondents. Table 1 displays summary statistics on the distribution of banks by the numbers of respondents and correspondents. By far the most common number of correspondents is two, although nearly 2,000 banks report only one. For banks in our sample that are listed as correspondents by other banks (i.e., banks with respondents), 43% have only one respondent, and most have fewer than six. There are a number of banks, however, with large numbers of respondents, with 11 banks that served over 300 respondents each (all located in New York City). The pyramidal structure of correspondent relationships, partially caused by rules for national banks allowing them to hold fractions of their reserves in reserve and central reserve cities, created

TABLE 1

Summary Statistics for Numbers of Correspondents and Respondents in 1893

Number of correspondents	Count	Number of respondents	Count
1	1,935	1	593
2	9,179	2–5	397
3	510	6–20	221
4	34	21–100	115
5	7	101–300	27
6	1	> 300	11
Total	11,666	Total	1,364
Average per Bank	2.03	Average per Bank	16.04

a concentration of interbank deposits in New York, making its clearinghouse the *de facto* central actor during financial crises (see Gorton and Tallman 2018). This structure meant that during a crisis, respondent banks might be unable to access their interbank deposits as upstream reserve agents shored up their own balances.

The Panic posed a unique challenge to the correspondent system in that, unlike other panics of the National Banking era, it originated outside of New York City and resulted in far more bank suspensions (500 banks suspended between May and August; the Panic of 1873 takes second place with 101 total bank failures). Two major economic developments in the early 1890s contributed to the onset of the Panic: first, an increase in business and commercial failures and an accompanying decline in economic activity; and second, concerns about the ability of the U.S. Treasury to maintain its commitment to the gold standard in the face of declining gold reserves. Contemporary observers noted the importance of both factors, with Sprague (1910), for example, emphasizing the former and Noyes (1909) stressing the latter. More recent work supports both factors as contributing to the start of Panic (see Carlson 2013). Wicker (2000, p. 58) points to a public statement by Treasury Secretary Carlisle on April 21 suggesting that the Treasury might need to begin redeeming its notes in silver rather than gold, presumably if the Treasury's gold reserves were to continue falling, as the effective start of disruption in the financial markets.

In the wake of these developments, a stock market crash on May 3 precipitated a series of bank failures, including several large banks in Chicago and one in New York. In June, banking panics began in the cities of Chicago, Omaha, Milwaukee, Los Angeles, San Diego, and Spokane. The NYCH stopped publishing bank-specific balance sheet information for its members on June 17, and began issuing clearinghouse loan certificates for settling interbank balances on June 21. In July, the Panic spread to Kansas City, Denver, Louisville, and Portland, and members of the NYCH partially suspended payments to the interior on August 3, generating a currency premium and currency hoarding. Bank suspensions continued, with more than 100 banks suspending in the month of August. The Panic ended when the currency premium vanished and the NYCH suspension was lifted in early September.

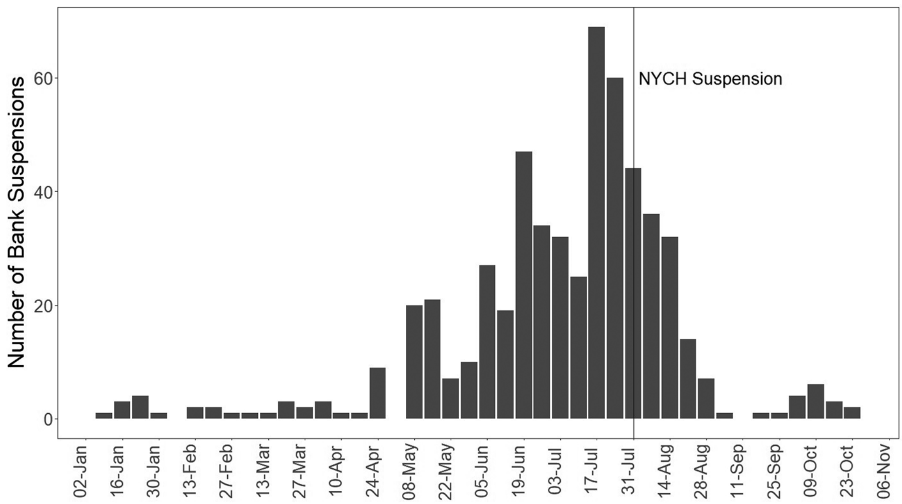

Fig 1. Weekly Bank Suspensions during the Panic of 1893.

Figure 1 shows the number of weekly bank suspensions from January through October.[1] The numbers are not unusual until the Treasury Secretary's statement in late April, with variations over a sharply upward trend in the ensuing weeks that rose to a crescendo in late July. Suspensions receded after the NYCH banks partially stopped cash payments to the interior on August 3, reaching very low levels by September, only to experience a final uptick in October. Overall, 76% of suspensions between April 21 and the end of October occurred before the partial suspension of the NYCH banks.

Table 2 sheds light on the nature and geographic distribution of the failures as well as their distribution across cities by reserve city status from May through October. Sixty-three percent of suspensions occurred in the *Bradstreet's* Western and Northwestern regions (what we would now call the Midwest and Upper Midwest/Plains), while another 29% occurred in the Southern and Pacific states and various territories of the time. The relatively small numbers of closures in New England and the Mid-Atlantic region (i.e., New York, New Jersey, Pennsylvania, and Delaware), where much of the nation's banking activity was concentrated, indicate that the Panic was indeed an event characterized by runs on banks in interior cities. Banks in the central reserve cities suspended at a rate of only 2.2%, compared to the 4.5% rate in other reserve cities and 4.2% among country banks. And while the absolute numbers of

1. Of the 584 bank suspensions listed in *Bradstreet's*, three do not have a given suspension date and an additional 24 do not appear in the 1893 *Banker's Directory*. This leaves our data set with 557 total bank suspensions in 1893, 531 of which occur after the April 21 breach of the gold reserve threshold.

TABLE 2

MONTHLY BANK SUSPENSIONS BY REGION AND RESERVE CITY STATUS IN 1893

	Region							Reserve city status		
	New Eng.	Mid-Atl.	West	Northwest	South	Pacific	Territories	Country	Reserve city	Central res. city
May	0	4	28	9	9	1	0	44	3	4
June	0	8	31	29	9	43	2	107	10	5
July	7	4	105	51	21	21	5	192	22	0
Aug.	2	9	27	43	29	2	1	105	6	2
Sept.	0	0	1	2	3	1	0	7	0	0
Oct.	4	1	4	0	5	0	1	15	0	0

NOTE: Regions follow *Bradstreet's* classification. Numbers include only those banks listed in *Bradstreet's* that could be linked to the Rand-McNally *Bankers' Directory*.
New England: Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont.
Middle Atlantic: New York, Delaware, New Jersey, and Pennsylvania.
Western: Colorado, Illinois, Indiana, Kansas, Kentucky, Michigan, Missouri, and Ohio.
Northwestern: Iowa, Minnesota, Montana, Nebraska, North Dakota, Wisconsin, and Wyoming.
Southern: Alabama, Arkansas, Georgia, Florida, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Virginia, and West Virginia.
Pacific: California, Idaho, Nevada, Oregon, and Washington.
Territories: Alaska, Arizona, Indian, New Mexico, Oklahoma, and Utah.
(South Dakota is not included as Bradstreet's was prevented from doing business there.)
Central reserve cities: New York, Chicago, and St. Louis.
Reserve cities: Albany, Baltimore, Boston, Brooklyn, Cincinnati, Cleveland, Des Moines, Detroit, Kansas City, Louisville, Lincoln, Milwaukee, Minneapolis, New Orleans, Omaha, Philadelphia, Pittsburgh, St. Joseph, St. Paul, San Francisco, and Washington, DC.

suspensions in reserve cities were relatively small, we will show that pressures on these banks led to severe problems for their downstream respondents.

We view the Panic as falling into two phases: a first consisting of bank runs in interior cities resulting in a drain of interbank deposits from New York reserve agents as described by Sprague (1910), Wicker (2000), and others, followed by a second characterized by the partial suspension of convertibility of deposits to cash in New York and a currency premium for banks in the interior. The second phase saw a proportionate reduction of suspensions in the Midwest and Pacific regions relative to other regions, while banks in the South saw an increasing share of the closures. We examine both phases separately and find that the dynamics of correspondent suspensions change after the suspension of the NYCH. We also find that, while the majority of bank suspensions represented permanent liquidations, almost 30% of suspending banks would go on to reopen by the end of October. We examine permanent liquidations and temporary suspensions separately to test the importance of the correspondent network for each.

2. CORRESPONDENT RELATIONSHIPS AND BANK FAILURES

We begin the empirical analysis by estimating probit models for predicting the likelihood of a given bank suspending based on prior suspensions of its correspondents and respondents, along with other potentially important bank-level factors. First, we predict bank suspensions using binary indicators for prior correspondent and respondent suspensions along with the shares of a bank's total correspondent and respondent networks that had previously gone into suspension. We are able to characterize completely the incidence of suspension in a given bank's downstream network because our data on primary correspondent relationships are comprehensive, covering the entire country. This allows us to identify all of a correspondent bank's respondents and thus analyze both upward and downward channels of transmission along the network. We next investigate whether the importance of either channel was affected by the partial suspension of payments of the NYCH banks. Finally, we separate temporary suspensions from permanent failures to determine whether correspondent and respondent suspensions were important for either or both. The results speak to whether correspondent and respondent suspensions primarily represented liquidity or solvency threats during the Panic.

2.1 *Correspondent and Respondent Suspensions*

Our first probit specifications, with estimates shown in Table 3, examine how the suspension of a given bank's upstream correspondent or downstream respondent relates to its own likelihood of suspension. The general setup is

$$\begin{aligned} Suspension_i = \ & \beta_1 Prior\,Correspondent/Respondent\,Suspension_i \\ & + \beta_2 Prior\,City\,Bank\,Suspensions_i + \beta_3 Clearinghouse\,Member_i \\ & + \beta_4 Clearinghouse\,City_i + \beta_5 Bank\,Type_i \\ & + \beta_6 \ln (Capital)_i + \beta_7 \ln (CountyPop)_i \\ & + \beta_8 Output\,Composition_i + \varepsilon_i, \end{aligned} \tag{1}$$

where the dependent variable is a binary indicator for whether or not a given bank suspended between April 21 and October 31, 1893. The variable of interest is a binary indicator for the prior suspension, over the same period, of at least one of the given bank's correspondents (columns (1) and (2)) or respondents (columns (5) and (6)), or the fraction of its direct correspondents (columns (3) and (4)) or respondents (columns (7) and (8)) that suspended prior to its own suspension.[2] We include the number of other banks in the city that suspended in the Panic to control for the possibility that observed effects of correspondent or respondent closures are related to location. For a bank that suspends, this measure excludes same-city suspensions that occur after the bank's own suspension. We also control for whether the given bank was a member of a clearinghouse, and whether or not it was a national bank. Membership in a clearinghouse is indicated in the *Bankers' Directory*, and we deduce whether a bank is a national bank from its name. We also include county-level controls from Haines (2010) for population and the industrial composition of the county, defined as the share of farm output in total output (i.e., the sum of farm and manufacturing). Columns (1)–(4) report estimates from specifications considering the role of

2. There is no relevant "prior" period for a given bank that does not suspend, so the explanatory variables for these banks are limited only by the sample period. We also omit banks with no recorded correspondents or respondents from the relevant estimations as they, by definition, cannot receive the treatment of interest.

TABLE 3

Bank Suspensions and the Role of Correspondents and Respondents

	All banks				Banks with respondents			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Correspondent suspended prior	0.053***	0.049***						
	(0.010)	(0.011)						
Fraction of correspondents suspended prior			0.067***	0.063***				
			(0.010)	(0.011)				
Respondent suspended prior					0.043*	0.049*		
					(0.026)	(0.030)		
Fraction of respondents suspended prior							0.077**	0.091**
							(0.034)	(0.038)
Prior city suspensions	0.010***	0.009***	0.010***	0.009***	0.008***	0.005*	0.009***	0.006**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.002)	(0.003)
ln(Capital)		0.008***		0.008***		0.012		0.016*
		(0.002)		(0.002)		(0.008)		(0.008)
Nonnational	0.011**	0.016***	0.010**	0.015***	0.007	0.020	0.003	0.016
	(0.004)	(0.005)	(0.004)	(0.005)	(0.015)	(0.017)	(0.015)	(0.017)
Clearinghouse member	−0.0005	−0.015	0.001	−0.015	−0.038*	−0.060**	−0.032*	−0.052*
	(0.010)	(0.010)	(0.010)	(0.010)	(0.021)	(0.030)	(0.020)	(0.028)
Clearinghouse city	0.026*	0.043**	0.027*	0.044**	0.060**	0.084**	0.059**	0.082**
	(0.014)	(0.020)	(0.014)	(0.020)	(0.029)	(0.037)	(0.029)	(0.038)
ln(County population)	−0.021***	−0.024***	−0.021***	−0.024***	−0.038***	−0.043***	−0.034***	−0.039***
	(0.002)	(0.003)	(0.002)	(0.003)	(0.007)	(0.008)	(0.007)	(0.008)
Ratio of farm output to total output	−0.050***	−0.049***	−0.050***	−0.049***	−0.070**	−0.072*	−0.060*	−0.058
	(0.008)	(0.009)	(0.008)	(0.009)	(0.035)	(0.037)	(0.034)	(0.037)
Mean of dependent variable	0.045	0.045	0.045	0.045	0.055	0.059	0.055	0.059
Pseudo R^2	0.06	0.07	0.06	0.07	0.12	0.12	0.11	0.11
Observations	11,666	9,699	11,666	9,699	1,364	1,182	1,364	1,182

Note: The table presents estimates of marginal effects from probit regressions of the incidence of bank suspension on selected covariates with robust standard errors in parentheses. Significance levels: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

correspondent suspensions in the full sample of all networked banks, while columns (5)–(8) examine suspensions after restricting the sample to banks with respondents. Selected columns include the bank's effective capital (defined as the sum of paid-in capital, surplus, and undivided profits) from the *Bankers' Directory*, which reduces the sample size when used by 17% in columns (1)–(4) and 13% in columns (5)–(8). All results report marginal effects with robust standard errors.

Columns (1) and (2) relate the prior suspension of a correspondent to about a 5 percentage point increase in the likelihood of suspension, while columns (3) and (4) relate a one-standard-deviation increase in the share of correspondents in suspension to a 1 percentage point increase in that likelihood (the vast majority of banks have exactly two correspondents, as per Table 1). Given that the baseline rate of suspension in the full sample is slightly over 4%, the suspension of a correspondent bank more than doubles the likelihood that a given bank will suspend. Columns (5)–(8) explore the impact of respondent suspensions and are limited to 1,364 banks. The results relate the prior suspension of a respondent to an increase of 4 to 5 percentage points in the correspondent's own probability of suspension, and relate a one-standard-deviation increase in the share of respondents in suspension to a 1 percentage point increase. Given a baseline rate of suspension in this sample of nearly 6%, having at least one respondent suspend nearly doubles a bank's own likelihood of suspension relative to a bank with no respondent suspensions. Recall from Table 1 that the average number of respondents in the sample is 16; thus, even a relatively small change in the fraction can mean several respondents in suspension.

For all specifications, we find that the number of prior bank suspensions within a given bank's city over the sample period is strongly associated with its own likelihood of closure. This is reasonable in that citywide runs were understood to be important drivers of bank suspensions. Even controlling for this factor, however, prior correspondent and respondent suspensions remain significant factors driving bank suspensions. Nonnational banks are more likely to suspend in the "all banks" regression, but not in the restricted sample of banks with respondents (60% of banks in this restricted sample are national banks). For the restricted sample, we also find that correspondent banks with membership in a clearinghouse are less likely to suspend. This is consistent with previous work on the proposed risk-pooling functions that clearinghouses took on during panics (Jaremski 2015). We also control for whether the bank was located in a city with a clearinghouse, for the simple reason that these represent many of the cities that experienced runs (e.g., Chicago, Milwaukee, Kansas City, and Denver). The inclusion of both variables makes it clear that, although cities with clearinghouses did not avoid bank runs, membership in a clearinghouse helped prevent a given bank's suspension. Being located in a populous county also decreases the incidence of suspension. This likely reflects the Panic's concentration in regions outside of New York and the Northeast.

The positive and often significant coefficient on a bank's effective capital requires explanation, as it is consistent with a counterintuitive inference that better capitalized banks were more likely to fail. Lacking another measure of a bank's size, however, capital primarily captures the effect of being a larger bank. Further, with covariates such as county population and being located in a clearinghouse city also included in the regressions, capital represents a bank's size conditional on other features of the bank's city or county. We likely see positive coefficient here because, conditional on these factors, larger banks would tend to be more interconnected with the rest of the banking system, and it is this connectedness that makes them more likely to suspend than less connected banks in similar locations. Given that we do not have other measures of a bank's connectedness for our comprehensive sample of banks, we see this effect manifesting through size. Considering the small effect of including capital on the main coefficients of interest (i.e., prior suspensions) and the reduction of the sample sizes, largely through the omission of smaller banks that are a key element of our contribution, we do not report results with the capital variable moving forward. Those results are available upon request.

We also explore whether, for banks serving as correspondents, the effect of respondent suspensions can be explained by the difference in severity of the Panic in the areas where their respondents are located. In other words, does the location of respondent banks matter, and does the impact of respondent suspensions vanish when liquidity shocks in their locations are considered? To test this, we construct a weighted average of the state-level decline in deposits in the states where a bank's respondents are located.[3] We include this alongside our measure of respondent suspension to examine whether locational severity among a bank's respondents matters independently of prior respondent suspensions. We find that banks exposed to respondents in states with larger deposit declines were more likely to suspend, even when controlling for a direct respondent suspension. This result is consistent with a role for liquidity shocks in transmitting the Panic throughout the network. We stop short of interpreting the result causally due to the timing of state deposit declines, which reflect failures occurring throughout the Panic and thus not necessarily those occurring prior to suspension of the correspondent of interest (i.e., on the left-hand side). Most importantly for our analysis, however, the impact of direct respondent suspensions remains statistically significant, indicating that our respondent suspension measure is not simply a proxy for locational exposures to the Panic.

To further check for robustness, we considered a number of other factors that may have affected bank suspensions and summarize the results here. For respondents, we found that prior suspension of an in-state correspondent had a smaller but still statistically significant effect on the likelihood of closure, suggesting that suspensions of out-of-state correspondents were more difficult to manage and thus more disruptive. As seen earlier in Table 1, our sample includes banks with a wide range of connections to other banks, and we might expect the importance of bank suspensions to depend

3. Specifically, we used percentage-point declines in the individual deposits of national banks at the state level from the *Annual Report of the Comptroller of the Currency* (1893, Vol. 1, pp. 277–327), as reported for March 6, 1893 (before the Panic) and October 3, 1893, to construct a weighted average of a given correspondent's exposure to respondents from different parts of the country, some of which may have been more disrupted by the Panic. We compute the shares of a given correspondent's total number of respondents in each state, and use these fractions as weights in computing overall exposure of that correspondent.

on the overall number of connections that a bank has. We find that the effect of a prior suspension of a correspondent on a given bank's own probability of suspension was smaller if it had multiple correspondents, presumably due to diversification across interbank partners, but remained statistically significant. Similarly, for correspondents we found that the effect of a respondent's suspension was less severe for banks with more respondents. For example, separate regressions for banks with three or more correspondents and for banks with fewer than three connections indicate that the primary driver of the results in Table 3 is the set of banks with few connections; for these banks, a related bank suspension is strongly significant, while the effect loses statistical significance for banks with many connections. This further supports our intuition that suspension of a correspondent/respondent bank is particularly impactful for banks with few other connections to draw upon.

The findings in this subsection reinforce the importance of the respondent network as a propagation mechanism for financial contagion during the Panic. Moreover, suspensions spread both up and down the correspondent network. This is consistent with Calomiris, Jaremski, and Wheelock (2022), who find that during the Great Depression both correspondent and respondent failures significantly increased the probability of bank suspension.

2.2 *The Impact of the NYCH Suspension*

The NYCH member banks partially suspended payments to interior banks on August 3, 1893, fundamentally changing the nature of the Panic. Prior to this, reserves held by New York banks had declined sharply as interior banks became increasingly concerned about depositor withdrawals and drew down their own deposits with reserve agents. The partial suspension of payments led many downstream banks that had been freely redeeming deposits to limit payouts to customers, and created a premium on currency that would last through August. While the first phase of the Panic was characterized by deteriorating banking conditions spreading from the interior up to banks in New York, now financial unrest spread from New York down to respondents unable to access their deposits with reserve agents. We now examine whether the character of bank suspensions changed with the suspension of payments by the NYCH banks.

To do this, we restrict our outcome variable to capture whether a suspension occurred before or after the NYCH event. The first two columns of Table 4 replicate our earlier specifications on the role of correspondent suspensions for all banks in the sample, but include only those suspensions occurring before August 3. Columns (3) and (4) include only bank closures following the suspension, leaving banks closing before August 3 out of the sample. (Of the 531 *Bradstreet's* suspensions in our data set, 127 occurred after August 3, while the other 404 suspended before.)

The results indicate that while the suspension of a correspondent was a significant predictor of bank closure prior to the suspension of payments by the NYCH, it ceased to be an important factor thereafter. This suggests that once all major banks in New York had partially suspended payments, respondents were more or less

TABLE 4

Bank Suspensions before and after the NYCH Suspension

	All banks				Banks with respondents			
	Pre-NYCH suspension		Post- NYCH suspension		Pre-NYCH suspension		Post-NYCH suspension	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Correspondent suspended prior	0.050***		0.005					
	(0.010)		(0.004)					
Fraction of correspondents suspended prior		0.059***		0.007				
		(0.008)		(0.006)				
Respondent suspended prior					0.034		0.012	
					(0.023)		(0.019)	
Fraction of respondents suspended prior						0.062*		0.020
						(0.032)		(0.014)
Prior city suspensions	0.007***	0.007***	0.003***	0.003***	0.005**	0.005**	0.004**	0.005**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Nonnational	0.008**	0.007*	0.003	0.002	0.003	−0.001	0.004	0.004
	(0.004)	(0.004)	(0.002)	(0.002)	(0.014)	(0.014)	(0.007)	(0.007)
Clearinghouse member	−0.005	−0.006	0.012	0.012	−0.039**	−0.034*	0.004	0.008
	(0.008)	(0.008)	(0.012)	(0.012)	(0.020)	(0.019)	(0.012)	(0.017)
Clearinghouse city	0.037***	0.038**	−0.008*	−0.008*	0.080***	0.080***	−0.033	−0.037
	(0.014)	(0.014)	(0.005)	(0.005)	(0.028)	(0.029)	(0.029)	(0.033)
ln(County population)	−0.019***	−0.019***	−0.002*	−0.002*	−0.033***	−0.030***	−0.005	−0.004
	(0.002)	(0.002)	(0.001)	(0.001)	(0.006)	(0.006)	(0.003)	(0.003)
Ratio of farm output to total output	−0.048***	−0.047***	−0.003	−0.003	−0.066*	−0.056*	−0.006	−0.005
	(0.007)	(0.007)	(0.004)	(0.004)	(0.035)	(0.033)	(0.013)	(0.014)
Mean of dependent variable	0.034	0.034	0.011	0.011	0.044	0.044	0.012	0.012
Pseudo R^2	0.07	0.07	0.03	0.03	0.12	0.12	0.28	0.28
Observations	11,666	11,666	11,270	11,270	1,364	1,364	1,304	1,304

Note: The table presents estimates of marginal effects from probit regressions of the incidence of bank suspension on selected covariates with robust standard errors in parentheses. Significance levels: $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

equally unable to utilize their relationships with correspondents effectively. Even in the absence of a prior correspondent suspension, banks were still unable to access their interbank deposits or to rely on their New York correspondents.[4] In columns (5)–(8), which consider only correspondent banks, we also find a significant effect of respondent suspensions prior to the NYCH suspension and insignificant effects thereafter.

Our main takeaway is that the suspension of the NYCH represented a severe disruption to the interbank network as a whole, and that in such an environment any particular linkages in the interbank network mattered less. Numerous banks still suspended after this event (the unconditional probability of failure in the postevent sample is 1.0% versus 3.4% before), but interbank correspondent and respondent relationships were no longer a significant determinant of which banks those turned out to be. In other words, the suspension of a correspondent or respondent increases a given bank's own suspension risk relative to other banks only when those others have well-functioning interbank relationships, which was not the case following the suspension of the NYCH.

2.3 Temporary versus Permanent Closures

Of the 531 bank suspensions included in our data set, 161 of these banks would reopen by the end of October (a rate of 31%). In this subsection, we investigate whether the effects of prior correspondent suspensions differ for banks that closed and subsequently reopened versus banks that ceased operations permanently. This distinction speaks to whether the suspension of a bank's correspondent represents a problem of liquidity or of solvency. Carlson (2005) outlines potential reasons why such a closure might contribute to either liquidity or solvency issues. The suspension of a correspondent bank could decrease a bank's liquidity by interfering with its access to deposits at that correspondent, but could also lead to solvency issues by inhibiting the bank's ability to clear checks and drafts. If the primary impact of a correspondent suspension is to create a liquidity problem for a given bank, we might expect banks suspending under this circumstance to be more likely to reopen once the Panic subsides. If the suspension of a correspondent represents a significant threat to a given bank's solvency, however, a permanent failure would be more likely.

Table 5 lists the numbers of suspending banks that eventually reopened or closed permanently. The reopening rate for banks suspending in May was only 17%, but the rates rose to about 30% in June and July, and to 44% in August. These figures seem

4. To examine this conjecture further, we checked whether having correspondents in Chicago and St. Louis (the other two designated central reserve cities in 1893) mitigated the effect of a prior suspension of a correspondent after the NYCH event, presumably because a given respondent would be more diversified in its interbank partners, and found the interaction not statistically significant. Similarly, we checked whether having more non-NYC correspondents mitigates the impact of a prior correspondent suspension after the NYCH event, and that interaction was also not statistically significant. These results likely reflect the ultimate connection of every bank to New York City. Even for banks with correspondents in other cities, those correspondents themselves likely depended strongly on New York, making the clearinghouse suspension truly a universal event for all banks, regardless of their direct connections to other locations.

TABLE 5

Temporary and Permanent Suspensions by Month

	Suspended and reopened	Closed permanently
May	9	42
June	38	84
July	66	148
Aug.	50	63
Sept.	2	5
Oct.	0	15

to reflect the increasing role of liquidity issues relative to solvency concerns as the Panic wore on, though both were clearly present.

Table 6 shows the impact of prior correspondent and respondent suspensions by the nature of the closure. As before, columns (1) and (2) confirm that the prior suspension of a correspondent increases a given bank's general probability of suspending, whether that be temporary or permanent. Evaluated at the means of the dependent variables, however, prior suspension of a correspondent renders a given bank 2.5 times more likely to close and eventually reopen, but only twice as likely to close permanently. Columns (3) and (4) consider the role of prior respondent suspensions in the failures of their correspondents. For this sample, limited to banks with respondents, suspension of a higher fraction of respondents does not affect the probability of a given correspondent to close and reopen, but does increase its likelihood of permanent closure. For example, a bank with 50% of its respondents suspending, evaluated at the mean of the closure rate in the sample, is nearly twice as likely to close permanently than a bank with no respondent failures.

Columns (5)–(8) focus only on suspended banks and examine whether, conditional on suspension, the prior suspensions of correspondents are associated with more or less severe outcomes, and thus are more likely to be liquidity or solvency events. Columns (5) and (6) use a binary indicator for whether a suspended bank would reopen as the dependent variable and demonstrate a strong positive relationship between correspondent suspension and eventual reopening. This demonstrates that bank suspensions related to suspended correspondents were more likely to be temporary, while those occurring for reasons unrelated to correspondent suspensions were more likely to be permanent. The finding supports the view that correspondent suspensions were primarily related to liquidity problems.

Columns (7) and (8) examine whether shareholders of respondent banks that closed after a correspondent suspension suffered smaller losses than those of other suspended banks. To build such as measure, which is similar to what we would today call Loss-Given-Default (LGD), we collected data on the fraction of proven claims by shareholders unable to be met for all national banks entering receivership after April 21, 1893 from the *Annual Report of the Comptroller of the Currency* (1894, pp. 274–89). Column (7) shows results from a linear model using raw LGD fractions

TABLE 6

CORRESPONDENT AND RESPONDENT SUSPENSIONS, TEMPORARY VS. PERMANENT CLOSURES, AND LOSSES TO EQUITYHOLDERS

	Probit (Marginal effects)				Probit (Marginal effects)		OLS	Ordered probit
	All banks		Banks with respondents		Suspended banks only			
	Suspended & Reopened	Permanent closure	Suspended & Reopened	Permanent closure	Reopened		LGD (National banks only)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Correspondent suspended prior	0.021***	0.032***			0.116**		−0.203**	−0.926**
	(0.006)	(0.008)			(0.055)		(0.087)	(0.455)
Fraction of respondents suspended prior			0.015	0.055**				
			(0.024)	(0.022)				
Fraction of correspondents suspended prior						0.166**		
						(0.078)		
Prior city suspensions	0.004***	0.005***	0.007***	0.002	0.033**	0.034**	−0.035*	−0.173**
	(0.001)	(0.001)	(0.002)	(0.002)	(0.015)	(0.015)	(0.019)	(0.086)
ln(Assets)					0.087***	0.089***	−0.078*	−0.264*
					(0.017)	(0.017)	(0.044)	(0.159)
Nonnational	−0.007**	0.017***	−0.003	0.009	−0.136***	−0.135***		
	(0.003)	(0.003)	(0.009)	(0.012)	(0.051)	(0.051)		
Clearinghouse member	0.008	−0.007	0.018	−0.039**	0.106	0.098	−0.121	−0.652
	(0.008)	(0.007)	(0.016)	(0.016)	(0.107)	(0.106)	(0.117)	(0.598)
Clearinghouse city	0.002	0.025*	−0.005	0.060*	−0.154*	−0.148*	0.292	1.257*
	(0.007)	(0.013)	(0.020)	(0.031)	(0.079)	(0.080)	(0.178)	(0.663)
ln(County population)	−0.007***	−0.014***	−0.018***	−0.017***	−0.019	−0.021	−0.073	−0.221
	(0.001)	(0.002)	(0.004)	(0.005)	(0.025)	(0.025)	(0.047)	(0.149)
Ratio of farm output to total output	−0.015***	−0.035***	−0.049**	−0.012	−0.051	0.056	−0.136	−0.478
	(0.005)	(0.007)	(0.025)	(0.025)	(0.088)	(0.088)	(0.131)	(0.467)
Mean of dependent variable	0.014	0.031	0.026	0.029	0.321	0.321	0.281	
R^2/Pseudo R^2	0.06	0.06	0.15	0.11	0.22	0.21	0.12	0.06
Observations	11,666	11,666	1,364	1,364	514	514	130	130

NOTE: Robust standard errors appear in parentheses beneath the coefficients. Significance levels: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The ordered probit has five LGD bins: 0–20% (75 obs.); 20−40% (6 obs.); 40−60% (8 obs.); 60−80% (20 obs.); and 80−100% (21 obs.)

on the left-hand side, and column (8) reports estimates from an ordered probit with five LGD categories representing increasing 20% loss increments. Both regressions indicate a strong negative relationship between prior correspondent suspension and the extent of shareholder losses given default. This further suggests that a correspondent suspension represented primarily a liquidity event for a given bank, as banks that suspended for this reason were likely to incur lower losses to shareholders than those suspending for other reasons.

3. BANKS IN THE NEW YORK CLEARINGHOUSE

Although the Panic was characterized by large numbers of bank failures outside of New York, within the city only one national bank and three state banks closed their doors, with one of the three reopening almost immediately. Thus, examining the impact of respondent suspensions on New York banks requires studying more than bank suspensions alone. To do this, we collect weekly bank balance sheet and stock price data from the *Commercial and Financial Chronicle*. The balance sheet data for the NYCH banks run from April 22 through June 10, 1893. Their publication was halted at that point to protect the weaker New York banks from being targeted for runs, yet 94 of the 531 bank suspensions in our regressions did occur during this period. Thus, these balance sheets still offer an opportunity to examine the impact of respondent suspensions on the well-being of New York banks early in the Panic. Given that the bulk of suspensions occurred after the NYCH ceased publication of balance sheet items, however, we use weekly stock prices of the New York banks over the entire Panic to test whether exposure to respondent suspensions was reflected in the public's valuations of these banks.

3.1 Weekly Balance Sheets of New York Banks, April 22 through June 10

We now estimate the relationship between respondent suspensions and the various balance sheet variables reported by NYCH banks from April 22 through June 10. Table 7 shows the impact of respondent suspensions on the logs of total loans, deposits, and specie reserves using a linear model in columns (1)–(3), and on their levels in millions of dollars using a Poisson model in columns (4)–(6).[5]

The specifications in Panel A use the number of respondents currently in suspension each week on the right-hand side for each New York bank. We also include bank-level fixed effects and time fixed effects by week. The bank-level fixed effects control for factors such as the size of the bank, the number of correspondents, and other characteristics that are fixed for each bank throughout the period, while the time effects allow us to capture the differential effect of respondent suspensions

5. Because our data on deposits, loans, and specie are slightly right-skewed, even when taking logs, we follow Cohn, Liu, and Wardlaw (2022) in estimating Poisson models with multiple fixed effects for robustness.

TABLE 7

NYCH Bank Balance Sheets and Respondent Suspensions

	Linear model			Poisson model		
	ln(Deposits) (1)	ln(Loans) (2)	ln(Specie) (3)	Deposits (4)	Loans (5)	Specie (6)
Panel A						
Respondents in suspension	−0.015*** (0.003)	−0.018*** (0.004)	−0.007 (0.005)	−0.167*** (0.031)	−0.177*** (0.028)	−0.013 (0.009)
Panel B						
Assets of respondents in suspension	−0.026*** (0.004)	−0.025*** (0.007)	−0.015* (0.008)	−0.301*** (0.043)	−0.260*** (0.046)	−0.025* (0.013)
Panel C						
Fraction of respondents in suspension	−0.320 (0.204)	−0.436*** (0.108)	−0.103 (0.439)	−1.677 (1.388)	−2.164*** (0.721)	0.243 (0.278)
Mean of dependent variable	7.25	7.04	1.19	7.25	7.04	1.19

Note: Balance sheet variables are in millions of dollars. Means of the dependent variables are in millions of dollars. All models include 456 observations on 57 banks, and include fixed effects for bank and for calendar weeks. Robust standard errors are in parentheses. Significance levels: $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

across New York banks while controlling for the aggregate progression of the Panic week by week. The high correlation of various balance sheet items with one another preclude their inclusion as control variables in the regressions. In Panel A, the results from the linear model suggest that the suspension of a respondent (or an additional respondent) is associated with a 1–2% decline in both deposits and loans at the New York bank in question. The Poisson model estimates these declines at about $167,000 for deposits and $177,000 for loans. Banks in the NYCH that were subject to respondent suspensions clearly fared worse during the early phase of the Panic than banks not subject to respondent suspensions.

The specifications in Panels B and C use the total assets of suspended respondents and the fraction of respondents in suspension for each NYCH bank, respectively, on the right-hand side. The results for the assets of suspended respondents reinforce the negative effects of suspensions on the loans and deposits of New York banks, and now also show a statistically significant decline in specie. When suspensions are measured as a fraction of the total respondent network in suspension, the estimates are generally less precise but still support the negative impact of respondent suspensions.

Taken together the findings suggest that suspensions in a NYCH bank's respondent network directly affected the strength of its balance sheet in the early part of the Panic. The contraction in lending by the New York banks also illustrates an important channel through which respondent suspensions likely impacted real economic activity throughout the Panic, which fits well with Mitchener and Richardson (2019), who find for the Great Depression that respondent suspensions led to declines in the deposits of reserve agents and a resulting contraction in lending and economic activity.

TABLE 8

NY Bank Stock Prices and Closed Respondents

	Dependent variable: Bank stock price	
	(1)	(2)
Respondent suspended	−0.016***	
	(0.004)	
Fraction of respondents in suspension		−0.165**
		(0.071)
Number of banks	65	65
Observations	1,226	1,226
Week fixed effects	Yes	Yes
Bank fixed effects	Yes	Yes

Note: Robust standard errors are in parentheses.Significance levels: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

3.2 *Weekly Stock Prices of New York Banks, April through October*

Although the NYCH ceased publishing bank-specific balance sheet information after June 10, as Rousseau (2011) stresses, bank stock prices in the National Banking period still reflected the public evaluation of how well different banks fared in the face of financial distress. Indeed, over 1893 stock prices fell for 53 of the listed New York banks while rising for only seven and remaining unchanged for nine. To capture these kinds of effects, the ordinary least squares specifications reported in Table 8 use weekly stock prices collected from the *Commercial and Financial Chronicle* for the period from April 29 to October 28 to examine the link between respondent suspensions and the market valuations of the New York banks. In the analysis, all bank stock prices are normalized, and both bid and ask prices are utilized to make best use of the available information.[6] Column (1) includes an indicator variable for whether the New York bank had a respondent that was currently in suspension during that week. Column (2) includes the fraction of respondents that were currently in suspension. Both specifications account for the possibility that suspended respondents could reopen as the Panic progressed, as some did. The results in column (1) relate the suspension of a respondent to a 1.6% decline in a given New York correspondent's stock price that week. The results with the fraction of suspended respondents in column (2) further support their negative impact on a bank's stock price. Once again, time and bank-specific fixed effects are included to ensure that the impact from respondent suspensions is capturing the differential effects of these events between banks and over time. Overall, the results strongly indicate that New York banks with a greater degree of respondent suspensions fared significantly worse in terms of market valuations during the Panic.

Taken together, the results in this subsection point to the suspension of respondent banks as a key channel for weakening the financial position of central reserve city

6. For bank-week combinations where both bid and ask prices are available, both are used. When only one is reported, it is used and given double weight in the regressions.

banks in New York. As the Panic developed and intensified, hundreds of banks in the interior began to suspend, and the New York banks most closely linked to these closures fared the worst. Such banks saw their deposits and specie decline significantly and reduced their lending as a result. As the Panic progressed through July and August, respondent suspensions continued to imperil the financial situation of their New York correspondents, as reflected in their share prices. Even without a wave of bank closures in New York City, the suspensions of respondent banks still had visible impacts on banking outcomes in the nation's financial center.

4. CONCLUSION

The network of correspondent banking relationships in the National Banking period offered a way for banks forbidden from branching to nonetheless settle payments made by draft or check, invest in financial securities, and borrow short term in major financial markets. In times of crisis, however, the network also became a primary conduit for transmitting financial stress throughout the system. During the Panic of 1893, suspensions of respondent banks in the interior put pressure on their correspondents, increasing their likelihood of closure and draining their reserves and deposits. Similarly, suspensions of correspondents threatened the ability of their downstream partners to access interbank deposits and meet the withdrawal demands of their customers. Correspondent suspensions contributed both to temporary as well as permanent closures, but were more closely associated with temporary suspensions, highlighting the role of liquidity. While prior suspensions of both upstream and downstream banks were important determinants of closures of their interbank partners before the partial suspension of the NYCH banks on August 3, they ceased to be significant factors thereafter. The latter finding may reflect broader dysfunction in interbank relationships following the NYCH suspension.

Despite the lack of bank closures in New York City, evidence from weekly balance sheets and from bank stock prices indicates that the failure of respondent banks did have a tangible effect on New York banks. Respondent suspensions contributed to the contraction of loans by New York banks that exacerbated the real effects of the Panic on the U.S. economy. Furthermore, stock prices indicate that equity investors knew of the risks faced by specific New York banks and were able to adjust their holdings as banks saw their downstream respondents suspend. These findings offer evidence that the U.S. interbank network, necessitated by the prohibitions on branch banking during this period, destabilized the banking system and contributed to the severity of banking crises. While the Federal Reserve by construction was intended to relieve systemic pressures by providing a lender of last resort, and federal deposit insurance was later instituted to avoid bank runs, the persistence of legislation aimed at protecting individual unit banks from competition with larger and better diversified multiunit banks kept the U.S. banking system prone to disruptions of both local and aggregate nature throughout the twentieth century, with some remnants of those policies remaining to this day.

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